

Srinath Ravi

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Education

- 2025-26 MS in Computer Vision, **Carnegie Mellon University** | Pittsburgh, USA **4.0/4.0**
Relevant Courses: Advanced Computer Vision, Visual Learning and Recognition, Learning for 3D, Intro to Robot Learning
- 2018-22 BE in Computer Science, **Dayananda Sagar College of Engineering** | Bengaluru, India **9.29/10**
Relevant Courses: Machine Learning, Data Structures with Applications, Software Engineering, Computer Networks, Database Management

Experience

Arcturus Studios, Research Intern

May 2026 - Aug 2026

- Accelerated 3D Gaussian Splatting initialization using monocular depth estimation and devised a novel frame-to-frame initialization reuse strategy, eliminating redundant COLMAP recomputation and cutting compute costs by 50% for multi-camera live captures.
- Fine-tuned depth estimation models on large-scale human action datasets to supply precise geometric priors, significantly improving point cloud generation and rendering fidelity for fast-paced athletic movements.
- Architected custom vision algorithms to detect and remove physical occlusions, such as fences and nets, restoring unobstructed views for multi-camera live stadium broadcasts.

UnscriptAI, AI Research Scientist

Oct 2024 - Jun 2025

- Engineered advanced talking head models using 3D vision representations, enabling the creation of highly realistic avatars videos for large-scale advertising campaigns with top brands in India. The proposed method minimizes reliance on lengthy training video captures during photo shoots by 80%, significantly reducing production costs.
- Led a team to design automated pipelines for creating custom avatars and short-form videos, using advanced image and video generation models to deliver high-quality content at scale.

Capital One, Consultant

May 2024 - Oct 2024

- Built an end-to-end pipeline to reconstruct immersive 3D vehicle models from a single 2-minute video, enabling scalable 3D visualization for Capital One's vehicle resale platform. This work was done in collaboration with IISc.
- Proposed methods to detect inconsistencies in COLMAP pose estimates by leveraging extrinsic camera pose information, enabling early issue identification, improved computational efficiency, and reduced resource consumption.
- Formulated and implemented variations of 3D representations for a diverse set of videos captured in real-world conditions, including varying lighting and mobile phone cameras.

Vision and AI Lab, Indian Institute of Science, Research Assistant

Aug 2022 - Oct 2024

- Designed and developed an Implicit NeRF model called Strata-NeRF that uses VQ-VAE for handling Stratified Scenes
- Played a key role in creating ChromaDistill, a Radiance Field Network capable of achieving 3D-consistent colorization.
- Enhanced the performance of 3D Gaussian Splatting (3DGS) through Turbo-GS, introducing faster training, improved densification via error-guided methods, and convergence-aware budget control for high-quality rendering.

Projects

RefineGS: Generalizing 3D Gaussian Splatting for Out of Distribution Views [Project Page]

- Designed a real-time inference pipeline by implementing Conditional Flow Matching (CFM) and a feature-caching mechanism, delivering high-fidelity refinement at 20.45 FPS (a 40x speedup over SOTA diffusion methods like Difx3D+).
- Engineered a robust multi-view architecture using a weight-sharing CroCo v2 ViT backbone with Rotary Positional Embeddings (RoPE) for precise spatial awareness, training on 10k+ real-world scenes to maximize model generalization.

Object Manipulation in Zip-NeRF

- Developed OM-NeRF, a four-step method for comprehensive 3D scene decomposition and manipulation.
- Leveraged a pre-trained segmentation model to distill semantic information, enabling the prediction of semantic masks.
- Decomposed objects and backgrounds within 3D scenes using semantic information, and performed foreground object manipulations (translation, rotation, and duplication).

Publications

Turbo-GS: Accelerating 3D Gaussian Fitting for High-Quality Radiance Fields

CVPR 2026 (Highlight)

Srinath R*, Tao Lu*, Ankit Dhiman*, Emre Arslan, Angela Xing, Yuanbo Xiangli, R Venkatesh Babu, Srinath Sridhar

UniC-Lift: Unified 3D Instance Segmentation via Contrastive learning

AAAI 2026

Srinath R*, Ankit Dhiman*, Jaswanth Reddy, Lokesh R Boregowda, R Venkatesh Babu

ChromaDistill: Colorizing Monochrome Radiance Fields with Knowledge Distillation

WACV 2025, AI3DCC ICCV 2023

Ankit Dhiman, Srinath R, Srinjay Sarkar, Lokesh R Boregowda, R Venkatesh Babu

Strata-NeRF: Neural Radiance Fields for Stratified Scenes

ICCV 2023

Ankit Dhiman, Srinath R, Harsh Rangwani, Rishubh Parihar, Lokesh R Boregowda, Srinath Sridhar, R Venkatesh Babu

Skills

Language Python, C/C++, Java, Dart

Deep Learning/Vision 3D Vision (NeRFs, Gaussian Splatting, Point Clouds, Meshes), Generative AI (Diffusion, Flow Matching, GANs), Multimodal ML, Transformers (ViTs), Self-Supervised Learning

Reviewer: ACM Multimedia 23 | ACML 23 | AI-ML Systems 23 | WACV 24, 25 | CVPR 24, 26 | ECCV 24 | ICLR 25 | ICCV 25 | TPAMI | NIPS 26

Achievements: Awarded Intern of the Month at Springworks. | Best Project Award in VLR course (CMU), competing against 60+ teams.